

IEEE 14-Bus GFL↔GFM Switching

Corrected full-state equations, automatic formation, and the 250-s chronology

10 August 2026

Abstract

This report describes the corrected project-owned model used for one six-state EMF6 synchronous generator (SG) and four switchable inverter-based resources (IBRs). The correction restores the physical controller ownership: grid-following (GFL) mode contains a phase-locked loop (PLL), while grid-forming (GFM) mode forms its angle through a virtual swing equation and contains no PLL. Both modes share the AC-filter current and dc-link states. The requested chronology uses $\Delta t = 0.1$ s and events at 20, 50, 85, 85.15, 110, and 145 s, with a diagnostic endpoint of 250 s. The latest generated accepted trajectory ends at 200.000 s. Its observed network-voltage range is 0.014062–2.406517 pu and its aggregate-frequency range is 55.390979–84.329308 Hz. These extrema are reported as results, not used as tuned acceptance bands. The maximum accepted residual is 9.9395×10^{-9} , the deepest step subdivision is 6, and the number of rejected domain trials is 0. The SG re-closed at 154.300 s. Staged release retained authenticated GFM resource(s), at bus(es), because an SG-online all-GFL equilibrium is small-signal unstable. Numerical completion and operational readiness are reported separately; raw excursions are never hidden or used to tune a gate.

1 Scope, evidence classes, and non-goals

The system base is $S_{base} = 100$ MVA and $f_0 = 60$ Hz. SG1 is connected at bus 1; IBR1–IBR4 are connected at buses 2, 3, 6, and 8. Network equations, power flow, equilibrium, small-signal analysis, and time-domain simulation use project-owned Base MATLAB code. The model choices are separated as follows.

Class	Use in this work
Source-defined / source-mapped	AC-filter, GFL PLL and PI loops, GFM virtual swing and voltage-amplitude dynamics, controller gains, and event chronology.
Case-defined	Bus/branch data, resource ratings and limits, initial P/Q schedule, E_{ref} , event times, and healthy per-bus voltage references.
Project-derived	Fixed dual-mode state superset, dc-source closure, mode-transfer map, lost-SG dispatch participation, equilibrium gauge, selector ranking, grid-relative SG synchronizer, and subdivision-history initialisation.
Numerical method	Newton solves, finite-difference Jacobians, one rotational quotient coordinate, implicit trapezoidal integration, and event-local backward-Euler half steps.

The run does not claim EMT accuracy, unbalanced-fault validity, or controller field readiness. In particular, a transient voltage excursion is disclosed rather than clipped, filtered, or used to retune the equations.

2 Corrected fixed state and input contracts

Every IBR stores one 20-state vector of constant dimension. For presentation, a fixed permutation groups the GFM/plant block before the GFL-control block:

$$x_{IBR} = \begin{bmatrix} x_{GFM} \\ x_{GFL} \end{bmatrix} = \Pi x_{runtime}, \quad (1)$$

$$x_{GFM} = \left[i_d, i_q, V_{dc}, \theta, \omega, E, \xi_{Vd}, \xi_{Vq}, \xi_{Id}^{gfm}, \xi_{Iq}^{gfm}, V_{d,del}^{gfm}, V_{q,del}^{gfm} \right]^T, \quad (2)$$

$$x_{GFL} = \left[\delta_{PLL}, \xi_{PLL}, \xi_P, \xi_Q, \xi_{Id}^{gfl}, \xi_{Iq}^{gfl}, V_{d,del}^{gfl}, V_{q,del}^{gfl} \right]^T. \quad (3)$$

Here Π is a reporting-only permutation: production keeps its existing ABI and no state is re-ordered in code. The first three entries of x_{GFM} are the common physical plant and remain active in either mode. GFL activates those three plus the eight-state x_{GFL} controller and freezes the nine GFM-only states. GFM activates all 12 entries of x_{GFM} and freezes x_{GFL} exactly. Hence GFL has 11 active states including its PLL, while GFM has 12 active states including its VSG angle, frequency, and voltage amplitude but no PLL.

Table 1: IBR state inventory: 20-state vector $x_{IBR} = [x_{GFM}^T x_{GFL}^T]^T$. Common plant (rows 1–3) stays active in both modes. GFM-only states (rows 4–12) freeze in GFL mode. GFL-only states (rows 13–20) freeze in GFM mode.

#	State	Meaning	GFL	GFM
<i>Common plant (active in both modes)</i>				
1	i_d	d-axis filter current	•	•
2	i_q	q-axis filter current	•	•
3	V_{dc}	dc-link voltage	•	•
<i>GFM-only (freeze in GFM→GFL)</i>				
4	θ	VSG angle	○	•
5	ω	VSG frequency deviation	○	•
6	E	external voltage reference	○	•
7	ξ_{Vd}	d-axis voltage-loop integrator	○	•
8	ξ_{Vq}	q-axis voltage-loop integrator	○	•
9	ξ_{Id}^{gfm}	d-axis inner current integrator	○	•
10	ξ_{Iq}^{gfm}	q-axis inner current integrator	○	•
11	$V_{d,del}^{\text{gfm}}$	d-axis command delay	○	•
12	$V_{q,del}^{\text{gfm}}$	q-axis command delay	○	•
<i>GFL-only (freeze in GFL→GFM)</i>				
13	δ_{PLL}	PLL angle	•	○
14	ξ_{PLL}	PLL integrator	•	○
15	ξ_P	active-power outer integrator	•	○
16	ξ_Q	reactive-power outer integrator	•	○
17	ξ_{Id}^{gfl}	d-axis inner current integrator	•	○
18	ξ_{Iq}^{gfl}	q-axis inner current integrator	•	○
19	$V_{d,del}^{\text{gfl}}$	d-axis command delay	•	○
20	$V_{q,del}^{\text{gfl}}$	q-axis command delay	•	○
<i>Total active states</i>			<i>11</i>	<i>12</i>

The input is

$$u = [P_{ref}, Q_{ref}, E_{ref}]^T. \quad (4)$$

GFL consumes P_{ref}, Q_{ref} . GFM consumes all three inputs; E_{ref} is an external terminal-voltage reference, not the measured $|V|$.

2.1 Per-unit boundary and terminal power

Let $\kappa = S_{base}/M_{base}$ and let ϑ be the active controller angle (δ_{PLL} in GFL or θ in GFM). The model uses the power-invariant project boundary

$$I_{sys} = \frac{(i_d + ji_q)e^{j\vartheta}}{\kappa}, \quad S_{sys} = P + jQ = VI_{sys}^*. \quad (5)$$

No isolated factor 3/2 is inserted in only part of the per-unit model. This identity is also the independent current/power oracle used by tests and reporting adapters.

3 Common converter plant and dc closure

With $Ve^{-j\theta} = v_d + jv_q$, limited current $i_d + ji_q$, filter parameters L_f, R_f , and per-unit frequency ω_c , the common current plant is

$$\frac{L_f}{\omega_b} \dot{i}_d = V_{d,del} - v_d - R_f i_d + \omega_c L_f i_q, \quad (6)$$

$$\frac{L_f}{\omega_b} \dot{i}_q = V_{q,del} - v_q - R_f i_q - \omega_c L_f i_d. \quad (7)$$

The circular limit acts on the controller current reference only. If $i^{*,raw} = i_d^{*,raw} + ji_q^{*,raw}$, then

$$i^* = \alpha i^{*,raw}, \quad \alpha = \min\left(1, \frac{I_{max}}{|i^{*,raw}|}\right). \quad (8)$$

The physical filter states i_d, i_q and the terminal-current injection are never projected, clipped, or replaced by i^* . Consequently a fast fault transient may drive actual filter current above the reference limit; the report discloses that response rather than altering the plant. The command delay is

$$\dot{V}_{d,del} = \frac{V_{d,cmd} - V_{d,del}}{T_d}, \quad \dot{V}_{q,del} = \frac{V_{q,cmd} - V_{q,del}}{T_d}. \quad (9)$$

The physical energy balance requires an external dc current, but the case does not prescribe its controller. The project therefore uses the declared exogenous controlled-current closure

$$I_{dc} = \frac{P_{ac}}{V_{dc}} + \frac{C_{dc}}{T_{dc}}(V_{dc,ref} - V_{dc}), \quad (10)$$

$$C_{dc} \dot{V}_{dc} = I_{dc} - \frac{P_{ac}}{V_{dc}}, \quad P_{ac} = V_{d,cmd} i_d + V_{q,cmd} i_q, \quad (11)$$

$$\therefore \dot{V}_{dc} = \frac{V_{dc,ref} - V_{dc}}{T_{dc}}. \quad (12)$$

Instantaneous power feed-forward prevents this unpublished closure from moving the sourced AC equilibrium. Nonpositive or nonfinite V_{dc} fails closed.

4 GFL equations: PLL present

In the 20-state vector (??), GFL mode activates the three common-plant states i_d, i_q, V_{dc} from x_{GFM} together with the eight-state controller x_{GFL} from (??), giving 11 active states, while it freezes the nine GFL-incompatible states ($\theta, \omega, E, \xi_{Vd}, \xi_{Vq}, \xi_{Id}^{gfm}, \xi_{Iq}^{gfm}, V_{d,del}^{gfm}, V_{q,del}^{gfm}$). The GFL frame is owned by its PLL:

$$v_d + jv_q = Ve^{-j\delta_{PLL}}, \quad (13)$$

$$\dot{\xi}_{PLL} = v_q, \quad (14)$$

$$\dot{\delta}_{PLL} = k_{p,PLL} v_q + k_{i,PLL} \xi_{PLL}, \quad (15)$$

$$\omega_{PLL} = 1 + \dot{\delta}_{PLL}/\omega_b. \quad (16)$$

Measured inverter-base power is $P_{inv} = \kappa P$ and $Q_{inv} = \kappa Q$. The outer loops are

$$e_P = \kappa P_{ref} - P_{inv}, \quad \dot{\xi}_P = e_P, \quad i_d^* = k_{pP} e_P + k_{iP} \xi_P, \quad (17)$$

$$e_Q = \kappa Q_{ref} - Q_{inv}, \quad \dot{\xi}_Q = e_Q, \quad i_q^* = -(k_{pQ} e_Q + k_{iQ} \xi_Q). \quad (18)$$

After circular limitation of $i_d^* + j i_q^*$, the inner current errors are $e_d = i_d^* - i_d$ and $e_q = i_q^* - i_q$, and

$$V_{d,cmd} = v_d + R_f i_d - \omega_{PLL} L_f i_q + k_{pI} e_d + k_{iI} \xi_{Id}, \quad (19)$$

$$V_{q,cmd} = v_q + R_f i_q + \omega_{PLL} L_f i_d + k_{pI} e_q + k_{iI} \xi_{Iq}. \quad (20)$$

Let $r_d = i_d^{*,raw} - i_d^*$ and $r_q = i_q^{*,raw} - i_q^*$. Anti-windup belongs to the outer P/Q integrators because their output is the limited current reference:

$$\dot{\xi}_P = \begin{cases} 0, & \alpha < 1 \text{ and } k_{iP} e_P r_d > 0, \\ e_P, & \text{otherwise,} \end{cases} \quad (21)$$

$$\dot{\xi}_Q = \begin{cases} 0, & \alpha < 1 \text{ and } (-k_{iQ}) e_Q r_q > 0, \\ e_Q, & \text{otherwise.} \end{cases} \quad (22)$$

The inner current controller has no voltage-command limiter in this model, hence $\dot{\xi}_{Id} = e_d$ and $\dot{\xi}_{Iq} = e_q$ without the outer-loop hold condition.

5 GFM equations: VSG angle, no PLL

In the 20-state vector (??), GFM mode activates all 12 states of x_{GFM} from Equation (??) and freezes the eight-state GFL controller x_{GFL} from Equation (??). GFM owns an absolute virtual-machine angle and frequency:

$$\dot{\theta} = \omega_b(\omega - 1), \quad (23)$$

$$M\dot{\omega} = \kappa P_{ref} - P_{inv} - D_v(\omega - 1), \quad (24)$$

$$\tau_E \dot{E} = k_Q(\kappa Q_{ref} - Q_{inv}) - k_E(E - E_{ref}). \quad (25)$$

Equation (25) is the decisive correction. Replacing E_{ref} by measured $|V|$ would remove the intended external voltage-restoration term at equilibrium and changes the small-signal physics. The voltage reference is

$$v_d^* = E, \quad v_q^* = 0, \quad qquad i_d^* = k_{pV}(v_d^* - v_d) + k_{iV} \xi_{Vd}, \quad i_q^* = k_{pV}(v_q^* - v_q) + k_{iV} \xi_{Vq}. \quad (26)$$

The limited current reference then enters the same physically signed inner current controller and command-delay plant. Here the limiter-owning states are the voltage-loop integrators:

$$\dot{\xi}_{Vd} = \begin{cases} 0, & \alpha < 1 \text{ and } k_{iV} e_{Vd} r_d > 0, \\ e_{Vd}, & \text{otherwise,} \end{cases} \quad \dot{\xi}_{Vq} = \begin{cases} 0, & \alpha < 1 \text{ and } k_{iV} e_{Vq} r_q > 0, \\ e_{Vq}, & \text{otherwise.} \end{cases} \quad (27)$$

The inner states still satisfy $\dot{\xi}_{Id} = e_d$, $\dot{\xi}_{Iq} = e_q$. No PLL state or hidden PLL angle is used in this branch.

6 Equilibrium, references, and SSSA

At SG trip the full nonlinear equilibrium solves every rectangular KCL row,

$$YV - I_{inj}(x, V, u) = 0, \quad (28)$$

plus one $\dot{E} = 0$ equation per selected GFM. Terminal GFM reactive power is a solved output; Q_{ref} and E_{ref} remain immutable case/event inputs. One scalar active-power mismatch is distributed by the frozen participation contract. A newly forming GFM holds its pre-trip P_{ref} when GFL resources remain; those GFL resources share the lost-SG deficit by P_{max} . If every IBR is GFM, the generic post-trip participation is used because no GFL remains.

One angle coordinate per energized island is removed as a mathematical gauge. This does not mean that only one physical GFM may operate. Several GFMs may form the same island, but they share one common rotational invariance. SSSA retains all relative electromechanical and controller modes and quotients only that common VSG-angle direction; no eigenvalue is deleted after calculation.

7 How the number of GFMs is selected

For SG-off operation the automatic selector enumerates $n_{GFM} = 1, 2, 3, 4$, every subset of that size, and every admissible reference member. Each candidate must pass topology, full-KCL equilibrium, operating limits, and the physical SSSA margin $\Omega \leq -\gamma_{req}$. The frozen ranking is: feasible first, fewer event-left mode changes, fewer GFMs, larger damping margin, deterministic resource-ID tie-break. Thus the current policy means *minimum small-signal-feasible formation*, not maximum transient robustness.

For the corrected case, local SSSA admits both a one-GFM point and the four-GFM point. The TDS answers the operationally relevant question: after SG trip the one-GFM formation did not provide enough large-disturbance support, so live J_V/J_f severity automatically raised the system to four GFMs. The report therefore presents the four-GFM spectrum below. The one-GFM and all candidate details remain audit artifacts, not presentation tables. This distinction is important: local small-signal feasibility is necessary evidence but is not proof of fault/load-step survival.

7.1 Complete selector and SSSA mode inventory

All 15 SG-off candidates remain generated and retained for audit. Failed candidates are never discarded or promoted to readiness, but the long candidate table is omitted from the report.

For a physical decision matrix A_p , right eigenvectors R and the paired left-eigenvector rows L satisfy

$$A_p R = R \Lambda, \quad L = R \backslash I, \quad \pi_{ki} = \frac{|R_{ki} L_{ik}|}{\sum_j |R_{ji} L_{ij}|}. \quad (29)$$

Participation is a reporting diagnostic and never feeds candidate ranking. Each complex-conjugate pair is printed once, with modal frequency $f_i = |\Im \lambda_i|/(2\pi)$ and damping ratio $\zeta_i = -\Re \lambda_i/|\lambda_i|$. For readability, each row shows only the mode number, eigenvalue, frequency, and the single largest-participation device:state. For repeated roots, especially the four exact -10 s^{-1} dc-link roots, individual eigenvectors may rotate inside the repeated eigenspace; they should be interpreted as one dc-link family rather than four uniquely attributable modes.

Disturbance-support point: four GFMs

Four 12-state GFMs give 48 full active states and 47 physical decision modes after the same one-angle quotient. The table has 30 rows with conjugate pairs printed once. No PLL state exists in this configuration. The limiting root is the real mode -2.344851 s^{-1} , dominated by

GFM voltage-loop integrators, while the principal VSG angle/frequency families appear at approximately 17.97–23.95 Hz.

No.	Eigenvalue λ (s ⁻¹)	f (Hz)	ζ	Dominant participation: device:state (normalised)
1	-3874.026967 j1062.858968	+ 169.159259	0.964364	IBR6: i_q (0.131); IBR6: i_d (0.131); IBR8: i_q (0.128)
2	-160.751647 j771.363375	+ 122.766294	0.204016	IBR6: i_q (0.306); IBR6: i_d (0.305); IBR3: i_q (0.108)
3	-159.870515	0	1.000000	IBR2: E^{GFM} (0.257); IBR8: E^{GFM} (0.254); IBR6: E^{GFM} (0.245)
4	-158.678405	0	1.000000	IBR8: E^{GFM} (0.709); IBR3: E^{GFM} (0.156); IBR2: E^{GFM} (0.108)
5	-158.281882	0	1.000000	IBR6: E^{GFM} (0.710); IBR3: E^{GFM} (0.199); IBR2: E^{GFM} (0.049)
6	-157.863132	0	1.000000	IBR2: E^{GFM} (0.573); IBR3: E^{GFM} (0.390); IBR6: E^{GFM} (0.018)
7	-142.863696 j1020.567960	+ 162.428436	0.138633	IBR8: i_q (0.343); IBR8: i_d (0.342); IBR3: i_q (0.061)
8	-118.069613 j555.888842	+ 88.472457	0.207763	IBR2: i_q (0.246); IBR2: i_d (0.245); IBR3: i_q (0.165)
9	-18.034003	0	1.000000	IBR8: ω_{VSG}^{GFM} (0.257); IBR6: ω_{VSG}^{GFM} (0.224); IBR2: ω_{VSG}^{GFM} (0.213)
10	-12.517483 + j16.868689	2.684735	0.595908	IBR2: $V_{d,del}^{GFM}$ (0.079); IBR3: $V_{q,del}^{GFM}$ (0.074); IBR2: $V_{q,del}^{GFM}$ (0.074)
11	-11.633833 + j13.272985	2.112461	0.659145	IBR3: ξ_{Iq}^{GFM} (0.087); IBR3: $V_{q,del}^{GFM}$ (0.087); IBR2: ξ_{Id}^{GFM} (0.084)
12	-10.000000	0	1.000000	IBR2: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
13	-10.000000	0	1.000000	IBR3: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
14	-10.000000	0	1.000000	IBR6: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
15	-10.000000	0	1.000000	IBR8: V_{dc} (1.000); IBR2: i_d (0.000); IBR2: i_q (0.000)
16	-8.750654 + j14.210463	2.261665	0.524347	IBR3: $V_{q,del}^{GFM}$ (0.176); IBR2: $V_{q,del}^{GFM}$ (0.170); IBR3: ξ_{Iq}^{GFM} (0.158)
17	-7.899977 + j14.415799	2.294346	0.480577	IBR8: $V_{q,del}^{GFM}$ (0.192); IBR8: ξ_{Iq}^{GFM} (0.166); IBR2: $V_{q,del}^{GFM}$ (0.051)
18	-7.522961 + j130.919399	20.836470	0.057368	IBR6: δ_{VSG}^{GFM} (0.353); IBR6: ω_{VSG}^{GFM} (0.299); IBR3: ω_{VSG}^{GFM} (0.096)
19	-7.491432 + j112.918894	17.971600	0.066198	IBR8: δ_{VSG}^{GFM} (0.409); IBR8: ω_{VSG}^{GFM} (0.299); IBR3: ω_{VSG}^{GFM} (0.078)
20	-6.973071 + j9.271486	1.475603	0.601073	IBR6: ξ_{Iq}^{GFM} (0.147); IBR6: ξ_{Id}^{GFM} (0.144); IBR6: $V_{q,del}^{GFM}$ (0.125)
21	-5.618627 + j17.153529	2.730069	0.311276	IBR6: $V_{q,del}^{GFM}$ (0.194); IBR6: ξ_{Iq}^{GFM} (0.137); IBR6: $V_{d,del}^{GFM}$ (0.104)

No.	Eigenvalue λ (s ⁻¹)	f (Hz)	ζ	Dominant participation: device:state (normalised)
22	$-4.987193 + j150.493385$	23.951766	0.033121	IBR3: δ_{VSG}^{GFM} (0.321); IBR2: ω_{VSG}^{GFM} (0.261); IBR3: ω_{VSG}^{GFM} (0.134)
23	$-4.598028 + j11.579746$	1.842974	0.369046	IBR8: $V_{d,del}^{GFM}$ (0.127); IBR8: ξ_{Id}^{GFM} (0.117); IBR8: $V_{q,del}^{GFM}$ (0.065)
24	-3.635015	0	1.000000	IBR2: ξ_{Vd}^{GFM} (0.362); IBR3: ξ_{Vd}^{GFM} (0.200); IBR3: ξ_{Vq}^{GFM} (0.117)
25	$-3.360497 + j0.551305$	0.087743	0.986809	IBR8: ξ_{Vq}^{GFM} (0.313); IBR8: ξ_{Vd}^{GFM} (0.244); IBR2: ξ_{Vd}^{GFM} (0.087)
26	-2.880396	0	1.000000	IBR2: ξ_{Vd}^{GFM} (0.325); IBR3: ξ_{Vq}^{GFM} (0.264); IBR2: ξ_{Vq}^{GFM} (0.223)
27	$-2.828050 + j1.409490$	0.224327	0.895001	IBR6: ξ_{Vq}^{GFM} (0.344); IBR6: ξ_{Vd}^{GFM} (0.300); IBR8: ξ_{Vd}^{GFM} (0.097)
28	$-2.503201 + j9.251750$	1.472462	0.261174	IBR2: $V_{d,del}^{GFM}$ (0.139); IBR2: ξ_{Id}^{GFM} (0.129); IBR3: δ_{VSG}^{GFM} (0.114)
29	-2.431408	0	1.000000	IBR2: ξ_{Vd}^{GFM} (0.395); IBR2: ξ_{Vq}^{GFM} (0.239); IBR3: ξ_{Vq}^{GFM} (0.179)
30	-2.344851	0	1.000000	IBR2: ξ_{Vq}^{GFM} (0.410); IBR2: ξ_{Vd}^{GFM} (0.184); IBR3: ξ_{Vd}^{GFM} (0.111)

7.2 SG-online 16-subset decision table

After SG1 is online, it owns the island's numerical angle gauge, but the four IBRs may still provide physical GFM support. All $2^4 = 16$ IBR mode subsets are retained in the evidence table; failed or unstable rows are not deleted. The decision eigenproblem first removes exactly one rigid network-angle coordinate, then calls eig; its finite-difference robustness check uses 0.5, 1, and 2 times the frozen Jacobian perturbation. The least-damped result across those three calculations is Ω_{worst} . A release is possible only when the exact one-IBR target row is authenticated and robustly stable.

The full 16-row SG-online classification remains an audit artifact rather than a report table. It explains why “one GFM or four GFMs” is not a manual binary choice. Four GFMs give distributed voltage-forming participation during the disturbance. Once SG1 is back and the live severity has recovered, the supervisor releases only one GFM at a time. It stops at the smallest stable subset supported by the table; it does not force all-GFL merely because SG1 is online.

8 Hybrid event and integration contract

At a mode transaction, the device reset map preserves the terminal port (V, I, P, Q) and V_{dc} , initializes only the target controller states, then solves the exact right-limit full-KCL algebraic system. A failure rolls the whole transaction back. Reference ownership is a numerical/supervisory gauge label, not a physical power injection; changing that label never replaces, removes, or changes a KCL row.

Normal continuous steps use coupled implicit trapezoidal integration:

$$x_{k+1} - x_k - \frac{h}{2} [f(x_k, y_k) + f(x_{k+1}, y_{k+1})] = 0, qquadg(x_{k+1}, y_{k+1}) = 0. \quad (30)$$

After a discontinuity, only the first interval is restarted with two backward-Euler half steps,

$$x_{m+1} - x_m - \frac{h}{2} f(x_{m+1}, y_{m+1}) = 0, \quad (31)$$

before returning to trapezoidal integration. This Rannacher restart damps the nonphysical step-to-step ringing of a stiff trapezoidal discretisation; it does not smooth, clip, or modify the physical trajectory.

If a logical output step requires dyadic subdivision, the next logical step starts from the most recently accepted subdivision level and then coarsens when the full step becomes feasible again. Every leaf still solves the identical implicit residual with the identical Newton tolerance. This history is only an initial step-partition choice: it changes neither the differential/algebraic equations nor the acceptance gate, and avoids repeatedly retrying coarse ancestors already known to fail in the fault and handback transients.

The per-device severity used for release contains exactly the requested voltage and frequency terms:

$$J_{V,i} = \frac{||V_i| - |V_{i,healthy}||}{0.10}, \quad J_f = \frac{|f_{COI} - 60|}{0.50}, \quad S_i = \text{sat}_{[0,1]}(0.5J_{V,i} + 0.5J_f). \quad (32)$$

The healthy voltage is the case-defined SG-online PF value at that IBR bus. RoCoF, active-power error, SCR, PLL-lock, and reference-availability diagnostics do not enter this scalar severity; applicable hard admissibility checks remain separate.

While the SG is offline the same two-term signal is evaluated in real time. If the system severity remains at or above $\Gamma_{on} = 0.65$ for $T_{d,on} = 0.10$ s, the selector commits the smallest authenticated feasible strict superset of the present GFM set. If it remains below $\Gamma_{off} = 0.35$ for $T_{d,off} = 1.00$ s, it may commit a feasible subset, but at least one GFM and one authenticated island reference are retained. In the generated run, SG trip initially committed one GFM and the measured stress added three resources at 21.2 s; this is an event-driven one-to-four transition, not a manual four-GFM schedule.

The breaker-open SG is synchronized to the live GFM-formed grid rather than to nominal zero slip. With $T_e = 0$, define

$$2H\dot{\omega}_{SG} = P_m - D\omega_{SG}, \quad \dot{\delta}_{SG} = \omega_0\omega_{SG}, \quad (33)$$

$$e_\theta = \text{wrap}(\angle V_{bus} - \angle V_{SG,oc}), \quad e_\omega = \omega_g - \omega_{SG}. \quad (34)$$

The project-derived mechanical command is

$$P_c = \text{sat}_{[0,P_{max}]} (D\omega_g + K_\omega e_\omega + K_\theta e_\theta), \quad (35)$$

$$K_\omega = 4H\zeta\omega_n - D, qquad K_\theta = \frac{2H\omega_n^2}{\omega_0}, \quad (36)$$

$$T_{sv}\dot{P}_{sv} = -P_{sv} + P_c, qquad T_{ch}\dot{P}_m = -P_m + P_{sv}, \quad (37)$$

with $\omega_n = 0.8$ rad/s, $\zeta = 1$, $T_{sv} = 0.2$ s, and $T_{ch} = 0.4$ s. The voltage matcher acts separately on field input. The close request at 145 s is only the earliest allowed time; the case provides a 20-s synchronization opportunity and retains the unchanged 0.5-s dwell. The breaker closed at 154.300 s with $\Delta V = 0.018675$ pu, $\Delta f = 0.000107$ pu, and $\Delta \theta = 5.612123^\circ$.